Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 30 (Marks: 1)

A page table needed for keeping track of pages of the page table is called ----------

Choices:

- 2-level paging
- Page directory
- Page size
- Page table size

TIME LEFT

79

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 20 (Marks: 1)

When drawing the resource allocation graph, processes are represented by --------and resources by ----------

Choices:

- Circles, Circles
- Circles, squares
- Squares, Circles
- Squares, Squares

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 8 (Marks: 1)

In paged segmentation, we divide every segment in a process into _____ pages.

Choices:

- Fixed size
- Variable size
- Small
- Large

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 32 (Marks: 1)

The critical section problem can be solved by the following except

Choices:

- Software based solution
- Firmware based solution
- Operating system based solution
- Hardware based solution

TIME LEFT

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 7 (Marks: 1)

Consider the case when multiple instances of a text editor are invoked. Only one copy of the editor needs to be kept in the physical memory. Each user's page table maps on to the same physical copy of the editor, but data pages are mapped onto different frames. This is the advantage of ----------.

Choices:

- Sharing in paging
- Inverse page table
- Sharing of segments
- Hashed page table

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 31 (Marks: 1)

The most important property of the working set is its _____.

Choices:

- Delay
- Thrashing
- Time
- Size

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 27 (Marks: 1)

---------holds the smallest legal physical memory address for a process

Choices:

- Stack pointers register
- Base register
- Limit register
- Index register

Question: 39 (Marks: 1)

_____ is a type of semaphore which is useful in multiprocessor system.

Choices:

- Spinlocks ←
- Mutual Exclusion
- Busy Waiting

Question: 40 (Marks: 1)

Following is NOT true about Virtual memory.

Choices:

- Virtual memory helps in executing bigger programs even greater in size that of main memory.
- Virtual memory makes the processes to stuck when the collective size of all the proce becomes greater than the size of main memory.
- Virtual memory also allows files and memory to be shared by several different proce through page sharing.
- Virtual memory makes the task of programming easier because the programmer nee worry about the amount of physical memory.

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 26 (Marks: 1)

_____ is used in the detection and recovery mechanism to handle deadlocks.

Choices:

- Wait-for Graph
- Resource allocation Graph
- Circular Graph
- Claim Edge Graph

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 50 (Marks: 5)

In Linux operating system, you are required to write down the working of copy on write technique with respect to virtual memory.

Answer:

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 49 (Marks: 5)

Describe the operations that a file owner can use for 'file protection'. Also mention s
operations that we can perform on a file.

Answer:

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 47 (Marks: 5)

Semaphores are used as synchronization tools for orderly execution of cooperating processes. However, sometimes incorrect use of semaphores may lead to Deadlock and starvation. You are required to differentiate between Deadlock and starvation with the help of code.

Answer:

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perating Systems (CS604)

Question: 34 (Marks: 1)

The statement "A process may hold allocated resources while waiting for of other esources" best describes which one of the following conditions?

Choices:

Mutual Exclusion

No preemption

Hold and wait

Circular wait

TIME LEFT
78

Type here to search

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Operating Systems (CS604)

Question: 44 (Marks: 3)

Consider a swapping system in which memory consists of the following hole sizes in memory order: 10KB, 4 KB, 20 KB, 18 KB, 7 KB, 9 KB, 12 KB, and 15 KB. Which holes is taken for successive segment requests of: (a) 12 KB (b) 10 KB (c) 9 KB For first fit?

Answer:

Operating Systems (CS604)

Question: 22 (Marks: 1)

If validation bit is 0, it indicates a/an --------- state of segment.

Choices:

- Protected
- Shared
- Legal
- Illegal

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 29 (Marks: 1)

Every ----------generated by the CPU is divided into two parts: a page number (p) and a page offset (d).

Choices:

- Page
- Process address space
- Physical address
- Logical address

TIME LEFT

80

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 46 (Marks: 3)

Suppose there's a deadlock occurs in the system and a deadlock detection algorithm detects a deadlock then how these deadlocks will be recovered? Mention any two method names.

Answer:

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 23 (Marks: 1)

The logical address of Intel 80386 is _____

Choices:

36 bits

48 bits

64 bits

128 bits

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 6 (Marks: 1)

Which of the following statements is false?

Choices:

- Virtual memory implements the translation of a program's address space into physical memory address space
- Virtual memory allows each program to exceed the size of the primary memory
- Virtual memory increases the degree of multiprogramming
- Virtual memory reduces the context switching overhead

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 17 (Marks: 1)

When there is no external fragmentation, free memory blocks are scattered on hard disk then _____ is used in order to utilize these blocks for space management.

Choices:

- Indexed Allocation
- Contiguous Allocation
- Linked Allocation
- Variable Allocation

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (CS604)

Question: 18 (Marks: 1)

The integer value of _____ semaphores can range over an unrestricted integer domain.

Choices:

Counting

Binary

Mutex

Bounded buffer

Operating Systems (CS604)

Question: 35 (Marks: 1)

A dashed line is used to represent a _____ in Resource Allocation Graph.

Choices:

- Claim edge
- Request edge
- Assignment edge
- Allocation edge

Question: 3 (Marks: 1)

In the _____algorithm the disk arm starts at one end of the disk, and moves toward the other end, servicing requests as it reaches each cylinder, until it gets to the other end of the disk

Choices:

- Scan
- SSTF
- Look
- C-Look

TIME LEFT

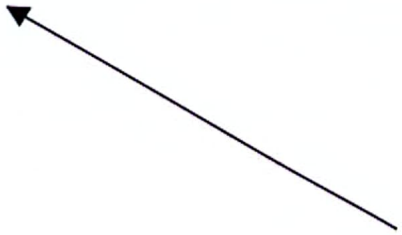
90

Question: 36 (Marks: 1)

You can use _____ to synchronize cooperating processes.

Choices:

- Semaphores



- Deadlocks
- Pipes

uestion: 40 (Marks: 1)

llowing is NOT true about Virtual memory.

.....

oices:

Virtual memory helps in executing bigger programs even greater in size that of main memory.



.....

Virtual memory makes the processes to stuck when the collective size of all the processes becomes greater than the size of main memory.

.....

Virtual memory also allows files and memory to be shared by several different processes through page sharing.

.....

Virtual memory makes the task of programming easier because the programmer need not worry about the amount of physical memory.

Question: 28 (Marks: 1)

External Fragmentation takes place in _____

Choices:

- Multiprogramming with Variable Tasks (MVT)
- Multiprogramming with Fixed Tasks (MFT)
- Multi-tasking
- Paging

Question: 13 (Marks: 1)

In the Scan algorithm, disk _____ starts at one end of the disk, and moves toward the other end, servicing requests as it reaches each cylinder, until it gets to the other end of the disk.

Choices:

Arm

Cylinder

Head

Vector

Question: 42 (Marks: 3)

Does pre-paging help to control the trashing? Justify your answer with solid argu

.....

Answer:

Question: 33 (Marks: 1)

2-process and n-process critical section problems are _____ solutions to solve the synchronization problems.

Choices:

- Software
- Hardware
- Operating system
- Firmware

Preventing the condition of _____ to happen, deadlocks can be prevented to happen.

Choices:

- Critical region
- Monitors
- Circular wait
- Critical section

TIME LEFT

82

Operating Systems (CS604)

Question: 12 (Marks: 1)

For page replacement algorithms, the page fault rate may increase as the number of allocated frames _____.

Choices:

- Keep Constant
- Increases
- Decreases
- Available

If members of a team have to work sequentially on the shared files then _____ is the most suitable scheme.

Choices:

- Common Login
- Common Group
- Link
- Duplicating Files

Question: 22 (Marks: 1)

If validation bit is 0, it indicates a/an --------- state of segment.

Choices:

- Protected
- Shared
- Legal
- Illegal

Question: 10 (Marks: 1)

----------is the separation of user logical memory from physical memory.

Choices:

- Virtual Memory
- RAM
- Physical memory
- ROM

Question: 37 (Marks: 1)

MMU stands for _____

Choices:

Major Management Unit

Memory Management Unit

Marginal Management Unit

Question: 45 (Marks: 3)

What are the issues that can be caused due to incorrect use of semaphores? You are required to write three issues.

.....

Answer:

Question: 24 (Marks: 1)

The term pager is used in connection with _____.

Choices:

- Demand Paging
- Paging
- Segmentation
- Paged Segmentation

Question: 41 (Marks: 3)

Explain the scenario where FIFO page replacement algorithm suffers from Belady's anomaly and how can you remove this anomaly?

Answer:

Question: 19 (Marks: 1)

With ---------you use condition variables.

Choices:

- Semaphores
- Read/Write Locks
- Swaps
- Monitor

Question: 4 (Marks: 1)

Assume a logical address space of 16 pages of 1024 words, each mapped into a physical memory of 32 frames. Each word consists of 2 bytes. What will be the total number of bits required for f (frames)?

Choices:

- 5
- 6
- 7
- 8

Question: 15 (Marks: 1)

ln –s command is used to create a _____link.

Choices:

- Soft 
- Hard
- Direct
- Indirect

Question: 25 (Marks: 1)

In a UNIX system, _____ system call can be used to request the operating system to memory map an opened file.

Choices:

- mmap( )
- fork ( )
- exec( )
- read( )

Question: 43 (Marks: 3)

Write a function that can swap two variables?

Answer:

Question: 14 (Marks: 1)

In case of thrashing, if CPU utilization is too low, the operating system _____ the degree of multiprogramming

Choices:

- Increases
- Decreases
- Increases or Decreases
- Keeps constant

Question: 16 (Marks: 1)

When a process needs to execute, memory is requested by the process. The size of the process is compared with the amount of adjacent memory available to execute the process. This method is called _____.

Choices:

- Contiguous Allocation
- Linked Allocation
- Indexed Allocation
- Variable Allocation

Question: 2 (Marks: 1)

_____correspond to character oriented devices, such as keyboard

.....

Choices:

- Directory

  .....

- Link File

  .....

- Ordinary File

  .....

- Character-special file

Question: 11 (Marks: 1)

In _____ scheme, number of frames allocated to a process is proportional to its size.

Choices:

- Propotional Allocation
- Priority Allocation
- Variance Allocation
- Fixed Allocation

Question: 9 (Marks: 1)

Consider process A is trying to access a page that does not belong to its address space. Which of the following protection scheme provide protection against such actions under paging?

Choices:

- Valid/invalid (v) bit
- Null bit/parity (n) bit
- Execute/parity (e) bit
- Write/null (w) bit

Question: 48 (Marks: 5)

There are high level and low level process synchronization constructs/tools. you need to enlist 5 constructs along with their type.

Answer:

Question: 5 (Marks: 1)

_____ occurs in a dynamic memory allocation system when many of the free blocks are too small to satisfy any request.

Choices:

- Segmentation
- Virtual memory
- Page fault
- Fragmentation

Question: 1 (Marks: 1)

_____ contains the names of other files and/or directories.

Choices:

- Block Special File
- Link File
- Ordinary File
- Directory

Question: 39 (Marks: 1)

_____ is a type of semaphore which is useful in multiprocessor system.

Choices:

- Spinlocks ←
- Mutual Exclusion
- Busy Waiting